Project: Worked examples
Title: Section properties
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Area, centroid, second moments of area (Ixx, Iyy, Ixy), principal axes and values, elastic and plastic section moduli, radii of gyration, torsion constant and (I-sections) warping constant for a parametric shape, an arbitrary polygon with holes, or a composite of placed parts. The x axis is horizontal (major axis of an I-section), y vertical.

WHAT IT CALCULATES	RELATION
Area and second moments of each outline, by Green's theorem	$A = \tfrac{1}{2}\oint(x\,dy - y\,dx)$; $I_{xx} = \int y^2\,dA$; $I_{yy} = \int x^2\,dA$; $I_{xy} = \int xy\,dA$
Second moments moved to the centroid by the parallel-axis theorem	$I_c = I - A\,d^2$
Principal axes	$\tan 2\theta = -2I_{xy}/(I_{xx} - I_{yy})$
Elastic section modulus	$W_{el} = I_{xx}/y_{max}$
Plastic section modulus, about the equal-area axis found by bisection	$W_{pl} = \lvert Q_{above}\rvert + \lvert Q_{below}\rvert$
Torsion constant by shape, a Saint-Venant series for a solid rectangle, $\pi d^4/32$ for a circle, Bredt for a closed thin-walled section, $\Sigma bt^3/3$ (plus a fillet term for rolled I) for an open section.	
Warping constant of an I-section	$I_w = I_{yy}\,(h - t_f)^2/4$

Basis: Properties of sections; torsion of solid and thin-walled sections[1](Appendix A, Table A.1; Ch. 10, Table 10.1); Torsional properties of rolled I-shapes[2](Appendix); Hot rolled steel channels, I and H sections: dimensions and masses[3]; Hot finished structural hollow sections: tolerances, dimensions and sectional properties[4].

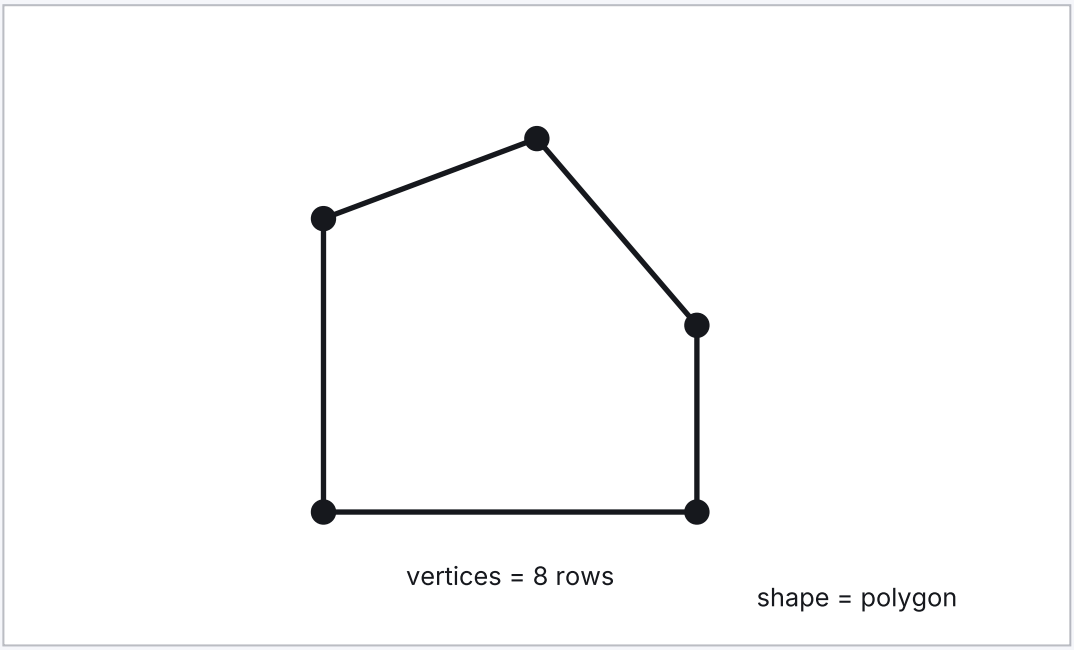


Figure 1 — A polygon through the vertices as listed (drawn representatively). Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
shape	Shape	polygon		drawing: DRG-BG-100 rev C	any
vertices	Polygon vertices (polygon shape)	8 row(s)		drawing: DRG-BG-100 rev C	any

TABLE ROW	VALUES
vertices[1]	x = -200 mm; y = -300 mm; loop = 1
vertices[2]	x = 200 mm; y = -300 mm; loop = 1
vertices[3]	x = 200 mm; y = 300 mm; loop = 1
vertices[4]	x = -200 mm; y = 300 mm; loop = 1
vertices[5]	x = -180 mm; y = -280 mm; loop = 2
vertices[6]	x = 180 mm; y = -280 mm; loop = 2
vertices[7]	x = 180 mm; y = 280 mm; loop = 2
vertices[8]	x = -180 mm; y = 280 mm; loop = 2

3. Assumptions

A1. A polygon outline is simple (no self-intersection); holes lie inside the outline and do not overlap. (Green's theorem integrates the signed loops as given.)
A2. Rolled-section root fillets are quarter circles of radius r; flange tips are square; hollow-section corners are quarter circles of outer radius r_o (default 1.5 t) and inner radius r_i (default 1.0 t). (Published section tables use the same idealisation.)
A3. Thin-walled torsion constants neglect the junction and tip effects except the fillet term kept for rolled I-sections. (Open sections: $\Sigma bt^3/3$; closed: Bredt.)

4. Working

[2] $A = \frac{1}{2} \oint (x\,dy - y\,dx)$ [1](Table A.1)

= Green's theorem over 8 vertices in 2 loop(s)

= 384 cm²

Area

[3] $x_c = \int x\,dA / A$

= Green's theorem over 8 vertices in 2 loop(s)

= 0 mm

Centroid x (from the shape origin)

[4] $y_c = \int y\,dA / A$

= Green's theorem over 8 vertices in 2 loop(s)

= 0 mm

Centroid y (from the shape origin)

[5] $I_{xx} = \int y^2\,dA - A\,y_c^2$

= Green's theorem over 8 vertices in 2 loop(s)

= 193150 cm⁴

Second moment of area about the centroidal x axis

[6] $I_{yy} = \int x^2\,dA - A\,x_c^2$

= Green's theorem over 8 vertices in 2 loop(s)

= 102270 cm⁴

Second moment of area about the centroidal y axis

[7] $I_{xy} = \int xy\,dA - A\,x_c\,y_c$

= Green's theorem over 8 vertices in 2 loop(s)

= 0 cm⁴

Product of inertia about the centroid

[8] $I_1 = (I_{xx} + I_{yy})/2 + \sqrt{(((I_{xx} - I_{yy})/2)^2 + I_{xy}^2)}$

= (193150 cm⁴ + 102270 cm⁴) / 2 + sqrt(((193150 cm⁴ - 102270 cm⁴) / 2)^2 + (0 cm⁴)^2)

= 193150 cm⁴

Major principal second moment

[9] $I_2 = (I_{xx} + I_{yy})/2 - \sqrt{(((I_{xx} - I_{yy})/2)^2 + I_{xy}^2)}$

= (193150 cm⁴ + 102270 cm⁴) / 2 - sqrt(((193150 cm⁴ - 102270 cm⁴) / 2)^2 + (0 cm⁴)^2)

= 102270 cm⁴

Minor principal second moment

[10] $\theta = \frac{1}{2}\,\mathrm{atan2}(-2\,I_{xy},\ I_{xx} - I_{yy})$

= ½ atan2(0, 90880)

= 0 deg

Angle from the x axis to the major principal axis (anticlockwise)

[11] $y_{top} = y_{max} - y_c$

= Green's theorem over 8 vertices in 2 loop(s)

= 300 mm

Extreme fibre above the centroid

[12] $y_{bot} = y_c - y_{min}$

= Green's theorem over 8 vertices in 2 loop(s)

= 300 mm

Extreme fibre below the centroid

[13] $x_{ext} = \max(x_{max} - x_c,\ x_c - x_{min})$

= Green's theorem over 8 vertices in 2 loop(s)

= 200 mm

Extreme fibre from the centroid, x

[14] $W_{el,x} = I_{xx} / \max(y_{top},\ y_{bot})$

= 193150 cm⁴ / max(300 mm, 300 mm)

= 6438.4 cm³

Elastic section modulus about x (governing fibre)

[15] $W_{el,y} = I_{yy} / x_{ext}$
$= 102270\ cm^4 / 200\ mm$
$= 5113.6\ cm^3$
Elastic section modulus about y
[16] $W_{pl,x} = |Q_{above}| + |Q_{below}|$ about the equal-area axis
= equal-area axis at y = 0 mm
$= 7776\ cm^3$
Plastic section modulus about x
[17] $W_{pl,y} = |Q_{right}| + |Q_{left}|$ about the equal-area axis
= equal-area axis at x = 0 mm
$= 5856\ cm^3$
Plastic section modulus about y
[18] $i_x = sqrt(I_{xx} / A)$
$= sqrt(193150\ cm^4 / 384\ cm^2)$
$= 22.428\ cm$
Radius of gyration about x
[19] $i_y = sqrt(I_{yy} / A)$
$= sqrt(102270\ cm^4 / 384\ cm^2)$
$= 16.32\ cm$
Radius of gyration about y
[20] $\alpha_{pl,x} = W_{pl,x} / W_{el,x}$
$= 7776\ cm^3 / 6438.4\ cm^3$
$= 1.2078$
Shape factor about x

5. Results

SYMBOL	VALUE	DESCRIPTION
A	384 cm²	Area
x_c	0 mm	Centroid x
y_c	0 mm	Centroid y
I_{xx}	193152 cm⁴	I_{xx}
I_{yy}	102272 cm⁴	I_{yy}
I_{xy}	0 cm⁴	I_{xy}
I_1	193152 cm⁴	I_1 (major principal)
I_2	102272 cm⁴	I_2 (minor principal)
θ	0°	Principal axis angle
$W_{el,x}$	6438.4 cm³	$W_{el,x}$
$W_{el,y}$	5113.6 cm³	$W_{el,y}$
$W_{pl,x}$	7776 cm³	$W_{pl,x}$
$W_{pl,y}$	5856 cm³	$W_{pl,y}$
i_x	22.43 cm	i_x
i_y	16.32 cm	i_y
$\alpha_{pl,x}$	1.2078	Shape factor, x

6. Checks (governing first)

None.

7. Applicability

WARNING J: no torsion constant is computed for an arbitrary polygon. [no-torsion]

8. Verification

Template struct.section-properties@1.1.1 Content hash 18d6536f1f7df3e1 Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. Young WC, Budynas RG, Sadegh AM. Roark's Formulas for Stress and Strain. 8th ed. McGraw-Hill; 2012.
2. Seaburg PA, Carter CJ. Torsional Analysis of Structural Steel Members. Steel Design Guide 9. American Institute of Steel Construction; 1997.
3. British Standards Institution. Hot Rolled Steel Channels, I and H Sections—Dimensions and Masses. BS EN 10365. BSI.
4. British Standards Institution. Hot Finished Steel Structural Hollow Sections—Part 2: Tolerances, Dimensions and Sectional Properties. BS EN 10210-2. BSI.